CS614 Gen AI with LLMs — Exhaustive Workload, Storage & Research Specification

Adaptive Multi-Hop RAG Project | Concrete Engineering Tasks, Local Storage Paths, and Master's Ablations

Architectural Rigor & Master's AI Standards: To satisfy rigorous graduate-level evaluation criteria, every member owns a distinct production engineering layer and an isolated empirical ablation study. All data and indexes are persisted locally in group_project/ (zero cloud storage fees, fully reproducible).

1. Summary Workload & Storage Matrix

Member	Engineering Ownership & Concrete Code	Local Storage & File Deliverable	Ablation Study (Master's Depth)	Rating
Member 1 (Anish)	LangGraph Orchestration, State DAG, Query Decomposer & Sufficiency Critic nodes.	src/orchestration/ agent_graph.py	Adaptive vs. Fixed-Hop Dynamics: Measures noise drift & premature stopping bounds.	**8.5/10**
Member 2	SEC EDGAR automated scraping, table preservation & 3x multi-format chunkers.	data/01_raw/ data/02_processed/ data/03_chunks/	Tabular Chunking Ablation: Compares Flat Text vs. Markdown Grids vs. JSON Key-Values.	**8.0/10**
Member 3	Dual indexing (ChromaDB + BM25), alpha score fusion, cross-encoder reranker.	storage/chroma_db/ storage/bm25.pkl	Hybrid Alpha Calibration & Reranker Cliff: Analyzes dense-sparse ratio & context depth.	**8.0/10**
Member 4	Structured JSON synthesis, citation mapper, automated regex factual verification node.	src/guardrails/ verifier.py	Guardrail Precision & Adversarial Stress: Benchmark on 20 conflicting/ambiguous disclosures.	**8.0/10**
Member 5	Automated RAGAS test harness, latency/token telemetry decorators, Streamlit dashboard.	results/scores.csv results/pareto.png app.py	Inference Economics & Model Tiering: Profiles Open 8B vs. Frontier models on Pareto curves.	**8.5/10**

2. In-Depth Member Engineering Specifications & Scientific Experiments

Member 1 (Anish): Adaptive Graph Orchestration & State Machine [Complexity: 8.5 / 10]

Engineering Responsibilities & Local Storage:

- *Storage Location:* group_project/src/orchestration/agent_graph.py, prompts/decomposer.yaml, prompts/critic.yaml.
- Constructs the LangGraph execution DAG: State schema holding query, sub_queries, accumulated_evidence, hop_count, and sufficiency_verdict.
- Implements query decomposition to break compound financial prompts into sequential atomic retrieval sub-goals.
- Builds the reflection Sufficiency Critic node to inspect accumulated context and conditionally trigger next hop or break to synthesis.

Dedicated Master's Ablation Study:

- *Adaptive Stopping vs. Forced Fixed-Hop Baseline:* Evaluates our dynamic critic against a rigid baseline forcing exactly 3 hops on every query. Measures retrieval drift (noise accumulation) and quantifies the exact accuracy degradation when extraneous context is retrieved.
- *Deliverable:* State-transition diagram, loop convergence rate table, and ablation plot of Accuracy vs. Hop Count.

Member 2: Data Ingestion, Table Extraction & Representation Chunking [Complexity: 8.0 / 10]

Engineering Responsibilities & Local Storage:

- *Storage Locations:* group_project/data/01_raw/ (raw 10-K HTMLs), data/02_processed/ (cleaned sections), data/03_chunks/ (RAG-ready chunk files).
- Writes automated Python scraper using sec-edgar-downloader to pull 2023-2024 filings for AAPL, MSFT, NVDA, TSLA, AMZN.
- Parses HTML hierarchies using BeautifulSoup/pdfplumber, isolating Item 7 (MD&A;), Item 7A (Market Risk), Item 8, and Footnotes with metadata tags.
- Builds 3 distinct chunking pipelines: (1) chunks_flat.jsonl (naive fixed-token sliding window), (2) chunks_markdown.jsonl (Markdown grid tables preserving headers), and (3) chunks_json.jsonl (Propositional key-value serialization).

Dedicated Master's Ablation Study:

- *Tabular Representation Ablation:* Benchmarks all 3 chunking formats on tabular multi-hop queries. Quantifies row/column misalignment rates, missing context rates, and year-swapping error rates (attributing 2023 figures to 2024).
- *Deliverable:* Comparative table extraction benchmark table and qualitative taxonomy of tabular representation failures.

Member 3: Dual-Index Retrieval, Alpha Fusion & Cross-Encoder Calibration [Complexity: 8.0 / 10]

Engineering Responsibilities & Local Storage:

- *Storage Locations:* group_project/storage/chroma_db/ (local Chroma SQLite & vector files), storage/bm25_index.pkl (serialized keyword index).
- Ingests chunks_markdown.jsonl from Member 2, embeds chunks with BAAI/bge-m3, and indexes them in ChromaDB.
- Tokenizes text and builds an in-memory BM25 index for exact matching of financial tickers, line items, and footnote numbers.
- Builds hybrid fusion formula: Score = $\alpha \cdot$ Dense + $(1 - \alpha) \cdot$ BM25, and integrates BAAI/bge-reranker-large to rescore top-25 merged pools down to top-5.

Dedicated Master's Ablation Study:

- *Hybrid Alpha Sensitivity & Context Cliff Analysis:* Evaluates α across {0.0, 0.3, 0.5, 0.7, 1.0} to prove why hybrid search outperforms pure dense vectors on alphanumeric financial tokens. Tests reranker compression depths (top-3 vs. top-5 vs. top-10) to map the context-cliff boundary.
- *Deliverable:* Sensitivity curve plotting Evidence Recall@5 vs. Hybrid α-weight and reranking context-depth performance chart.

Member 4: Factual Grounding, Citation Mapping & Hallucination Guardrails [Complexity: 8.0 / 10]

Engineering Responsibilities & Local Storage:

- *Storage Location:* group_project/src/guardrails/verifier.py, src/synthesis/generator.py.
- Designs structured generator prompt templates enforcing JSON output with inline citations mapped to exact source chunks and table IDs.
- Develops the post-generation Numerical Verifier node: extracts all dollar figures, dates, and percentages via regex/spaCy, asserting exact grounding against retrieved text.

Dedicated Master's Ablation Study:

- *Guardrail Efficacy & Adversarial Stress Testing:* Benchmarks system accuracy with Guardrails ON vs. OFF. Creates 20 adversarial queries with conflicting disclosures (e.g., overlapping guidance from 2023 vs 2024) to measure false-positive and false-negative rejection rates.
- *Deliverable:* Confusion matrix of hallucination detection, citation precision/recall curves, and qualitative case studies of intercepted hallucinations.

Member 5: Batch Evaluation Pipeline, Telemetry & Inference Economics [Complexity: 8.5 / 10]

Engineering Responsibilities & Local Storage:

- *Storage Locations:* group_project/results/scores.csv, results/pareto_frontier.png, app.py (Streamlit demo dashboard).
- Ingests FinQA/TAT-QA datasets and builds the automated evaluation script using the RAGAS library across 100 multi-hop queries.
- Instruments execution telemetry tracking latency (p50, p90, p99), input/output token counters, and API cost per query.
- Builds lightweight Streamlit web interface (app.py) enabling live side-by-side comparison of System A, B, and C with reasoning traces.

Dedicated Master's Ablation Study:

- *Inference Economics & Model Tiering:* Benchmarks 3 model tiers: (1) All-Frontier (GPT-4o/Claude), (2) Tiered Hybrid (Open-weight 8B for planner/retrieval, Frontier for synthesis), and (3) 100% Open-weight (Llama-3.1-8B). Plots empirical Pareto Frontier curves.
- *Deliverable:* Master comparative scorecard (Sys A vs B vs C) and Pareto curves plotting Faithfulness vs. Latency and Cost.